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DataCenter.lean — one tick as a distance and latency budget (exploratory).

Question: what does GateReadsLedger (the gate reads the ledger every tick) REQUIRE physically? A read is a round trip. The ledger must be close enough that a signal can go there and back inside 17 500 ps.

Constants (stated, integer, in picoseconds and micrometres):

c in vacuum 299 792 458 m/s
 signal in fibre $\sim c / 1.468$ (single-mode, $n \approx 1.468$) $\approx 204\,000$ km/s
 signal in copper $\sim 0.7\,c$ (PCB trace / twinax) $\approx 210\,000$ km/s
 core at 3 GHz one cycle = 333 ps $\rightarrow$ 52 cycles per tick
 typical latencies L1 ≈ 1 ns, L2 ≈ 4 ns, L3 ≈ 12 ns, DRAM ≈ 80 ns,
 NIC-to-NIC same rack $\approx 2\,\mu\text{s}$, cross-AZ ≈ 1 ms

Latency figures are order-of-magnitude industry values, stated as constants; the theorems are arithmetic on them. Core Lean 4, zero axioms.

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namespace A5O.DC

def periodPs : Nat := 17500

/-! ## 1. One-way and round-trip reach per tick, in micrometres. -/

def um_per_s_vacuum : Nat := 299792458 * 10 ^ 6

def um_per_s_fibre : Nat := 204200000 * 10 ^ 6

def um_per_s_copper : Nat := 210000000 * 10 ^ 6

def reach_um (v : Nat) : Nat := v * periodPs / 10 ^ 12 -- one way, per tick

def roundtrip_um (v : Nat) : Nat := reach_um v / 2 -- there and back

theorem vacuum_one_way : reach_um um_per_s_vacuum = 5246368 := by decide -- 5.25 m

theorem fibre_one_way : reach_um um_per_s_fibre = 3573500 := by decide -- 3.57 m

theorem copper_one_way : reach_um um_per_s_copper = 3675000 := by decide -- 3.68 m

theorem fibre_round_trip : roundtrip_um um_per_s_fibre = 1786750 := by decide -- 1.79 m

/-- The ledger a gate reads every tick over fibre must sit within 1.79 m of the gate: the same rack, at most. Not the next row. Not the network. -/

theorem ledger_must_be_within_two_metres :

roundtrip_um um_per_s_fibre < 2 * 10 ^ 6 := by decide

/-! ## 2. Distances in ticks. A replica d metres away is stale by $\geq$ this many ticks. -/

```
def ticks_one_way (d_um v : Nat) : Nat := (d_um * 10 ^ 12 + v * periodPs - 1) / (v * periodPs) --
ceiling
```

```
theorem next_rack_600mm : ticks_one_way 600000 um_per_s_copper = 1 := by decide
theorem across_hall_100m : ticks_one_way 100000000 um_per_s_fibre = 28 := by decide
theorem cross_campus_2km : ticks_one_way 2000000000 um_per_s_fibre = 560 := by
decide
```

```
/-- 1 ms cross-AZ latency, in ticks: fifty-seven thousand. -/
def ps_to_ticks_ceil (ps : Nat) : Nat := (ps + periodPs - 1) / periodPs
theorem cross_az_1ms : ps_to_ticks_ceil (10 ^ 9) = 57143 := by decide
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/-! ## 3. The core: 52 cycles per tick at 3 GHz. -/
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def cyclePs_3GHz : Nat := 333 -- 1/3 ns, floored
theorem cycles_per_tick_3GHz : periodPs / cyclePs_3GHz = 52 := by decide
```

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/-! ## 4. Memory: where the ledger snapshot can live if it is read every tick. -/
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def L1_ps : Nat := 1000
def L2_ps : Nat := 4000
def L3_ps : Nat := 12000
def DRAM_ps : Nat := 80000
def sameRackNIC_ps : Nat := 2000000
```

```
theorem L1_fits : L1_ps < periodPs := by decide
theorem L2_fits : L2_ps < periodPs := by decide
theorem L3_fits : L3_ps < periodPs := by decide
theorem DRAM_does_not : periodPs < DRAM_ps := by decide
theorem DRAM_in_ticks : ps_to_ticks_ceil DRAM_ps = 5 := by decide
theorem NIC_in_ticks : ps_to_ticks_ceil sameRackNIC_ps = 115 := by decide
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/-- Physical content of GateReadsLedger at one tick: the revocation snapshot
lives in on-die cache (L1–L3), not DRAM and not on a network hop. A
gate whose ledger is one DRAM access away is a stalePass 5; one NIC hop
away, stalePass 115. Those are the exposure windows ClockStandard.lean
proves for a stale gate. -/
```

```
theorem staleness_hierarchy :
  ps_to_ticks_ceil L3_ps = 1 ∧ ps_to_ticks_ceil DRAM_ps = 5 ∧
  ps_to_ticks_ceil sameRackNIC_ps = 115 ∧ ps_to_ticks_ceil (10 ^ 9) = 57143 := by
decide
```

```
end A5O.DC
```

```
#print axioms A5O.DC.vacuum_one_way  
#print axioms A5O.DC.fibre_round_trip  
#print axioms A5O.DC.ledger_must_be_within_two_metres  
#print axioms A5O.DC.across_hall_100m  
#print axioms A5O.DC.cross_az_1ms  
#print axioms A5O.DC.cycles_per_tick_3GHz  
#print axioms A5O.DC.DRAM_does_not  
#print axioms A5O.DC.staleness_hierarchy
```